

Can a Higher Shell Imprint the Acoustic Ruler? Feeding the S^4/S^5 Mode Ladders Through the BAO, and the Nested-Shell Beat

TZPID Gold Spine Series II, Paper XIV of XX

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Abstract

Paper I established that the baryon acoustic oscillation (BAO) ruler is the flat spherical-cavity spectrum: nodes are zeros of the spherical Bessel j_ℓ , with the first at $kr_s = \pi$. Paper XIII gave the curved spectra of the nested shells, $k_\ell R = \sqrt{\ell(\ell+n-1)}$. This paper feeds those curved $S^3/S^4/S^5$ ladders through the same acoustic test: can a higher or outer shell project a band into the observed ~ 150 Mpc ruler? The answer is conclusive and falsifiable. Dimension shifts the offset, not the spacing: at high ℓ every ladder steps by $1/R$, so a higher shell makes the same comb, shifted. With the Spartan curvature bound $|\Omega_K| = 0.005$ ($R_c = 62.9$ Gpc), the closed-enclosure mode comb is $\sim 2700\times$ finer than the BAO wiggle, so curvature is invisible at the acoustic scale—the mode-space twin of the permitted-but-unseen closed universe of Paper XII. A direct outer-shell imprint would require a ~ 23 Mpc cluster-scale shell, ruled out. Exactly one falsifiable channel survives: a nested-shell beat, in which two concentric shells differing in radius by a fractional $\varepsilon \approx 3.7 \times 10^{-4}$ produce a moiré comb on the BAO scale—possible but fine-tuned. Dimension's only genuine fingerprint is amplitude, not spacing: the degeneracy growth $\sim \ell^{n-1}$ deposits more power per band in higher shells, an envelope tilt rather than a new peak. The higher shells are neither refuted nor detected; the spine carries one specific, currently-null-but-not-excluded prediction.

1 Introduction: the one place a hidden shell could show

Paper XIII fixed the observable spatial slice at S^3 and relegated S^4 , S^5 to outer shells. If those shells are physical, can they leave any trace? The most sensitive standing-wave observable in cosmology is the BAO ruler, known to fractions of a percent. If a higher shell rings, that spacing is where it would show. We extend the worked BAO check of Paper I from the flat cavity to the curved nested ladders and find a clean, falsifiable answer.

2 Two ladders

The flat spherical-cavity acoustic eigenvalues are zeros of j_ℓ ; the $\ell = 0$ ladder is exactly harmonic, j_0 zeros $= \pi, 2\pi, 3\pi, \dots$ (Paper I). The curved enclosure on S^n has the Laplace spectrum

$$k_\ell R = \sqrt{\ell(\ell+n-1)} : \quad S^3 : 0, 1.73, 2.83, 3.87; \quad S^4 : 0, 2.00, 3.16, 4.24; \quad S^5 : 0, 2.24, 3.46, 4.58. \quad (1)$$

Remark 2.1 (Dimension shifts the offset, not the spacing). At high ℓ , $\sqrt{\ell(\ell+n-1)} \rightarrow \ell + (n-1)/2$, so consecutive modes differ by 1 in v for all n ; in physical wavenumber the spacing is $\Delta k \rightarrow 1/R$ independent of dimension. A higher shell makes the same comb, shifted.

3 Decisive test: curvature cannot shift the BAO

The BAO wiggle period is $\Delta k_{\text{BAO}} = 2\pi/r_s = 0.0427 \text{ Mpc}^{-1}$. The closed-enclosure mode comb has spacing $\Delta k_{\text{mode}} = 1/R_c$. With $|\Omega_K| = 0.005$, $R_c = c/(H_0\sqrt{|\Omega_K|}) = 62.9 \text{ Gpc}$, so $\Delta k_{\text{mode}} = 1.6 \times 10^{-5} \text{ Mpc}^{-1}$ and

$$\boxed{\frac{\Delta k_{\text{BAO}}}{\Delta k_{\text{mode}}} = \frac{2\pi R_c}{r_s} \approx 2700}. \quad (2)$$

Proposition 3.1 (Curvature is acoustically invisible). *The hyperspherical mode comb is $\sim 2700\times$ finer than the BAO wiggle; at acoustic scales it is effectively continuous, and the curved S^3 enclosure reproduces the flat j_ℓ ruler. This is the mode-space counterpart of Paper XII's result that a closed universe survives the distance data precisely because its discreteness cannot be seen at $|\Omega_K| \lesssim 0.005$.*

4 No direct outer-shell band

For a shell's ladder to land at the BAO period requires $1/R = \Delta k_{\text{BAO}}$, i.e. $R \approx 23 \text{ Mpc}$, galaxy-cluster scale, not a cosmic enclosure. Any outer shell ($R \gg \text{Gpc}$) has a ladder far too dense to imprint at $\sim 150 \text{ Mpc}$. There is no direct outer-shell BAO band.

5 The one falsifiable channel: a nested-shell beat

Two concentric shells of radii R and $R(1+\epsilon)$ have ladders that beat with comb spacing $\Delta_{\text{beat}} \approx 1/(R\epsilon)$. For the beat to fall on the BAO scale with $R = R_c$,

$$\boxed{\epsilon = \frac{1/\Delta k_{\text{BAO}}}{R_c} \approx 3.7 \times 10^{-4}} \quad (3)$$

so two nested shells differing by $\sim 0.037\%$ in radius beat into the BAO wiggle.

Remark 5.1 (A real but fine-tuned prediction). This is the single falsifiable channel by which the nest could imprint the ruler: a faint secondary periodicity locked to the BAO. It requires commensurability of the shell radii to one part in a few thousand, so it is a signature to search for, not a generic expectation—a hypothesis with a number attached, not a detection.

6 Dimension's true fingerprint: amplitude, not spacing

Dimension changes the degeneracy (power per band), not the spacing: $S^3 : 4, 9, 16, 25, 36 (\sim \ell^2)$; $S^4 : 5, 14, 30, 55, 91 (\sim \ell^3)$; $S^5 : 6, 20, 50, 105, 196 (\sim \ell^4)$. A higher shell pours more power per band—an amplitude or spectral-tilt signature in the acoustic envelope, never a new peak. A different and harder measurement than the ruler spacing.

7 Demarcation

Established physics: spherical-cavity acoustics and the j_ℓ ruler; the curved-space Laplace spectrum $\sqrt{\ell(\ell+n-1)}$; the BAO wiggle period $2\pi/r_s$; the moiré-beat relation $\Delta_{\text{beat}} \approx 1/(R\varepsilon)$.

Framework synthesis: the reading of the nested-shell ladders as physical acoustic spectra of outer enclosures; the conclusion that curvature and higher-shell dimension cannot shift the BAO spacing; and the identification of the nested-shell beat as the spine’s one BAO-observable prediction.

Predictions: (i) the nested-shell beat at $\varepsilon \approx 3.7 \times 10^{-4}$, to be searched for in high-precision power spectra; (ii) a degeneracy-driven amplitude or tilt in the acoustic envelope if higher shells contribute; (iii) the null result itself—no shift in BAO spacing from curvature at $|\Omega_K| \lesssim 0.005$ —which the framework predicts and present data confirm.

8 Limits of the construction

The analysis is at the level of mode spacings and degeneracies, not a full transfer-function computation; a complete treatment would propagate the nested modes through the photon–baryon Boltzmann hierarchy. The beat channel, while falsifiable, requires a coincidence the framework does not independently motivate. The entire analysis presumes the outer shells are physical resonators (Paper XIII); if they are mere embeddings, the beat channel is empty. What the paper establishes firmly is the negative result: at the measured curvature bound, no higher shell can move the BAO ruler’s spacing.

9 Conclusion: neither refuted nor detected

Fed through the BAO, the S^4/S^5 ladders leave the acoustic ruler untouched: their combs are either far too dense or would require cluster-scale shells. The ruler at ~ 150 Mpc remains an S^3 signal. The nest could reveal itself only through a fine-tuned beat at $\varepsilon \approx 3.7 \times 10^{-4}$ or a degeneracy-driven amplitude tilt—a single, specific, currently-null-but-not-excluded prediction. The series now turns to the operator algebra (Paper XV) before carrying the dark-energy, unification, and release-facing closure through Papers XVI–XX.

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